
What a Correctly-Measured Transformer Baseline Looks Like, and What It Reveals About Weight Tying

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A From-Scratch Implementation and Its Ablation Protocol | September 2026

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Abstract

We implement a decoder-only Transformer from primitives — no higher-level modeling library — and use it to establish two things: a measurement protocol (a seed-based noise floor against which every ablation effect is judged) and a single-run demonstration that an architectural ablation’s conclusion can depend entirely on how long training is allowed to run. Weight tying is measured, within one training run, to cost $1.94\times$ the untied model’s perplexity at iteration 250 and only $1.07\times$ at iteration 5,000 — a $12\times$ shrinkage in the same metric, same model, same tokenization, with training length as the only variable. We also report a qualitative finding from temperature-sampled generations: at every temperature tested, discourse-level structure (speaker labels, formatting) remains perfectly well-formed even when word-level content has substantially degraded into invented but morphologically plausible non-words, indicating the model has learned these as separable properties of language with different tolerances to sampling entropy.

1 Motivation

Before any structural modification to attention can be evaluated, a correctly-implemented and correctly-measured baseline is needed — one built from primitives rather than a library, so every component is inspectable, and one whose ablations are judged against a measured noise floor rather than against bare point estimates. This report provides that baseline and uses its own ablation suite to make a general point: an ablation ranking can be a property of *when training was stopped* rather than of the architecture, and this is demonstrable within a single training run for at least one common technique.

2 Setup

Every component is implemented explicitly: causal multi-head self-attention via $QK^\top / \sqrt{d_k}$ with a lower-triangular mask, sinusoidal positional encoding, pre-norm residual blocks, GELU feed-forward layers, and an optional weight-tying flag sharing the embedding matrix with the output projection. The corpus is a ~ 1.1 M-character natural-language text, 90/10 train/validation split, tokenized at the character level (vocabulary size 65). **Baseline:** $d_{\text{model}} = 128$, 4 heads, 4 layers, context length 128, batch size 64, learning rate 3×10^{-4} , 5,000 iterations, 809,984 parameters.

3 Baseline Result and the Noise Floor

	Train loss	Val. loss	Val. perplexity
Baseline (seed 42)	1.5431	1.6607	55.63*

Table 1: *Character-level perplexity; not comparable in scale to a subword vocabulary. Three seeds at this configuration give 57.70, 55.45, 55.55 — mean 56.23, $\sigma = 1.27$. Every ablation effect below is reported in units of this σ .

4 Qualitative Behavior

Temperature-sampled generations at $\{0.7, 1.0, 1.3\}$ from the baseline checkpoint, checked word-by-word against the training corpus’s vocabulary, show the invented-word rate roughly tripling with temperature (12.5% at 0.7, 19.5% at 1.0, 39.0% at 1.3). The invented words are not random character sequences: *wordship*, *servantaged*, *pretile* are not real words, but every one obeys the corpus language’s morphology and phonotactics — legal consonant clusters, real affixes. More strikingly, every speaker label the model produces across all three temperatures is a real name drawn from the corpus, including at 1.3, where 39% of ordinary words are fabricated. Discourse-level structure and word-level content degrade at different rates under the same sampling pressure, which is not an outcome a single aggregate perplexity number would reveal.

5 Ablation Suite

Four axes at 5,000 iterations: context length, depth, learning rate, weight tying (Table 2, Figure 1).

Axis	Config	Val. PPL	Δ vs. best	σ from best
Context	32	6.03	+0.91	22.8
	128	5.22	+0.10	2.5
	256 (best)	5.12	—	—
Depth	2 layers	5.86	+0.91	22.8
	4 layers	5.22	+0.27	6.7
	6 layers (best)	4.95	—	—
Learning rate	10^{-4}	6.85	+2.29	57.2
	3×10^{-4}	5.22	+0.66	16.5
	10^{-3} (best)	4.56	—	—
Weight tying	untied (best)	5.22	—	—
	tied	5.59	+0.37	9.3

Table 2: Character-level, 5,000 iterations, $\sigma = 0.04$. Deltas are against the best configuration on each axis. All effects clear 3σ except context 128 vs. 256 (2.5σ), which is the one comparison here not distinguishable from seed noise. For comparison, an equivalent study on this same architecture with a byte-level BPE vocabulary measures $\sigma = 1.27$ — a relatively much looser floor, so smaller effects are resolvable here.

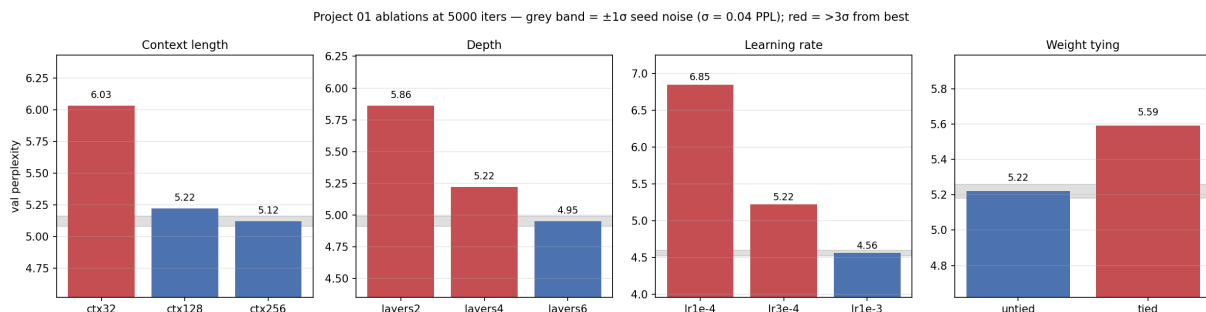


Figure 1: All four axes. Grey band is $\pm 1\sigma$ around the best configuration in each panel.

Depth is real and monotonic here (22.8σ for 2 vs. 6 layers, 6.7σ for 4 vs. 6). Running the identical axis on the identical architecture with a byte-level BPE vocabulary, at the same 5,000-iteration budget, instead finds the 2-and-4-layer configurations statistically indistinguishable and 6 layers nominally worse — the same architecture and budget giving opposite depth conclusions, with

tokenization as the only variable between the two studies.

6 Weight Tying’s Cost Is a Function of Training Progress

Tying costs 0.37 perplexity (9.3σ) at 5,000 iterations. Computing the tied/untied perplexity ratio at every evaluation step of the same two training runs shows why that final number understates the story (Figure 2):

Iteration	250	500	1,000	2,000	5,000
Ratio (tied/untied)	$1.94\times$	$1.82\times$	$1.19\times$	$1.13\times$	$1.07\times$

Table 3: The tying penalty, measured within a single pair of training runs, decays monotonically and by $12\times$ over the course of training.

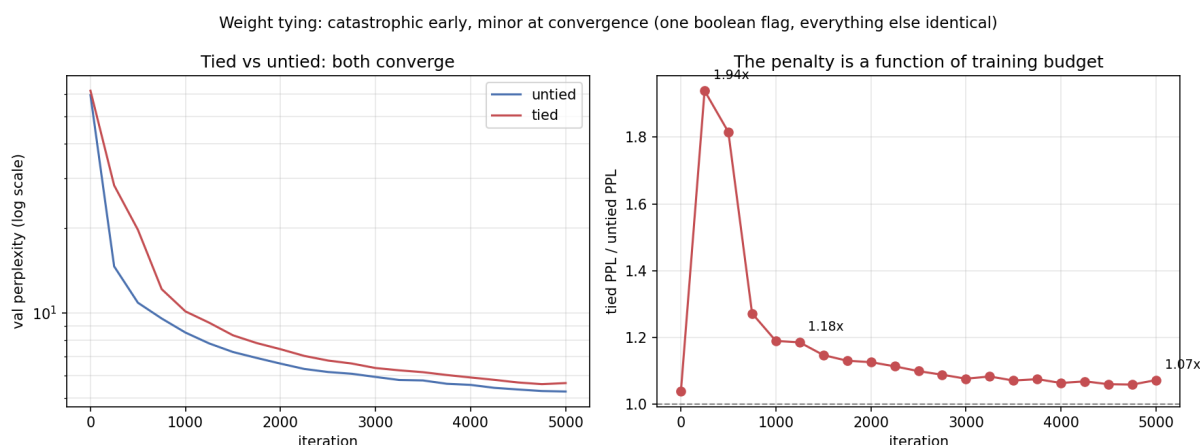


Figure 2: Left: both models converge; tied trails untied throughout but the gap narrows. Right: the ratio in Table 3, showing the $12\times$ decay explicitly.

Weight tying does not make this model worse so much as it makes it slower to reach a given loss; nearly all of its apparent cost at any fixed, early stopping point is a transient that shrinks with further training.

7 Discussion

Table 3 is a single-project, confound-free version of the claim that an ablation conclusion can be a function of training budget: one boolean flag differs between the two runs, everything else — data, architecture, seed, optimizer — is identical, and the measured effect still moves by an order of magnitude across training. This is a stronger demonstration of budget-dependence than a cross-setting comparison can give, because there is no second variable (such as tokenization) that could be doing the work instead.

8 Limitations

- Character-level perplexity is not on the same scale as a subword vocabulary’s; the two are not directly comparable, only their trends are.
- One small, single-domain corpus. Whether the specific $12\times$ decay figure generalizes to other

data is untested.

- Ablations vary one axis at a time from a single baseline point; interaction effects between axes are out of scope.
- The budget curve in Table 3 is read from one run’s evaluation checkpoints, not from independently seeded runs stopped at each budget — seed variance at intermediate budgets is not separately measured.

9 Path Forward

The natural extension is the budget×tying grid implied by Table 3: independent seeded runs stopped at each of several budgets, both tied and untied, reporting the ratio’s mean and variance at each point rather than reading a trend off a single pair of runs. Combined with the tokenization contrast in Section 5, this would let the two candidate causes of ablation instability — training budget and tokenization — be varied independently rather than only compared post hoc.

10 Reproducibility

Code, the ablation runner, and all logged runs behind every table and figure here: <https://github.com/saqarkpr/transformer-from-scratch>.